

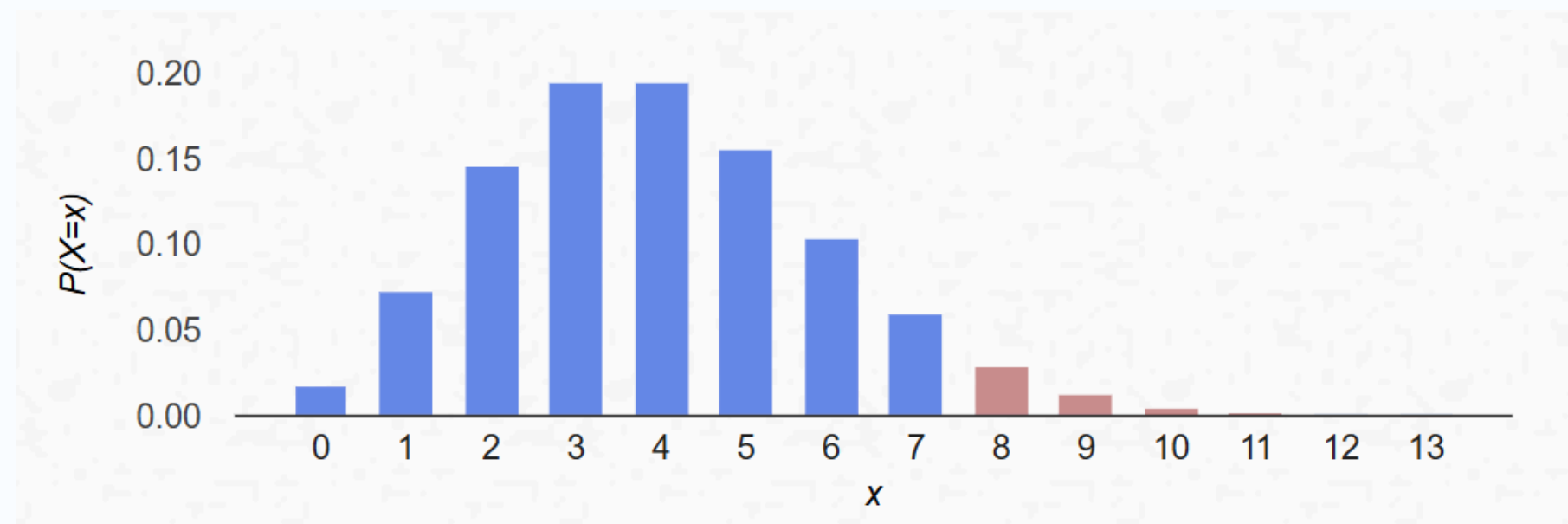
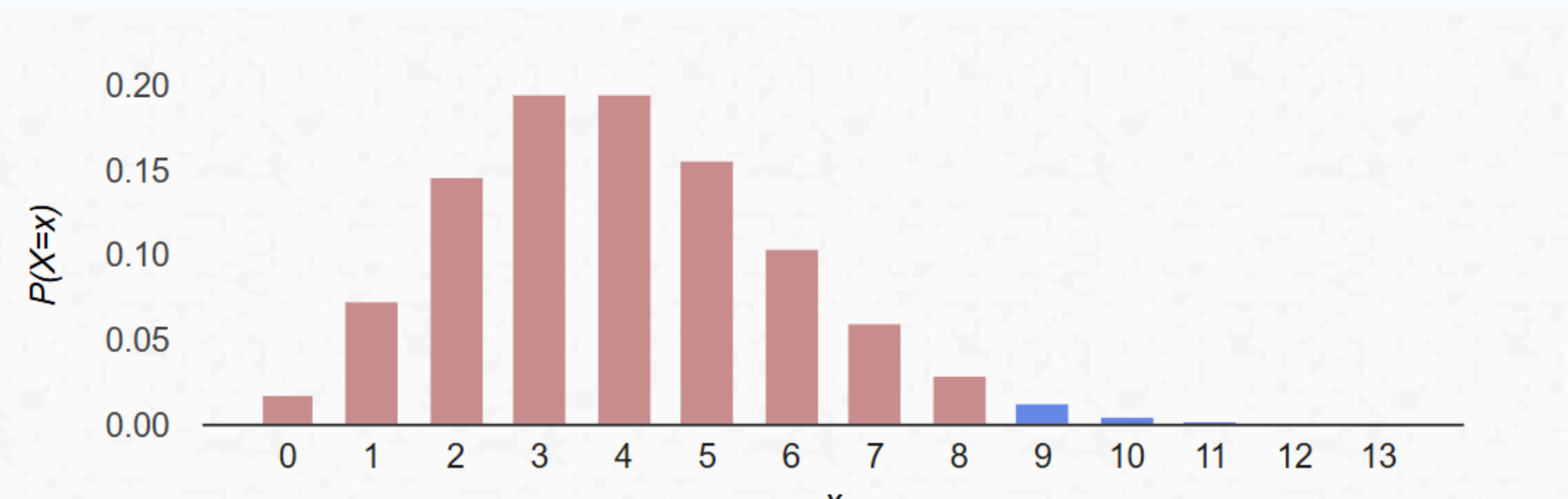
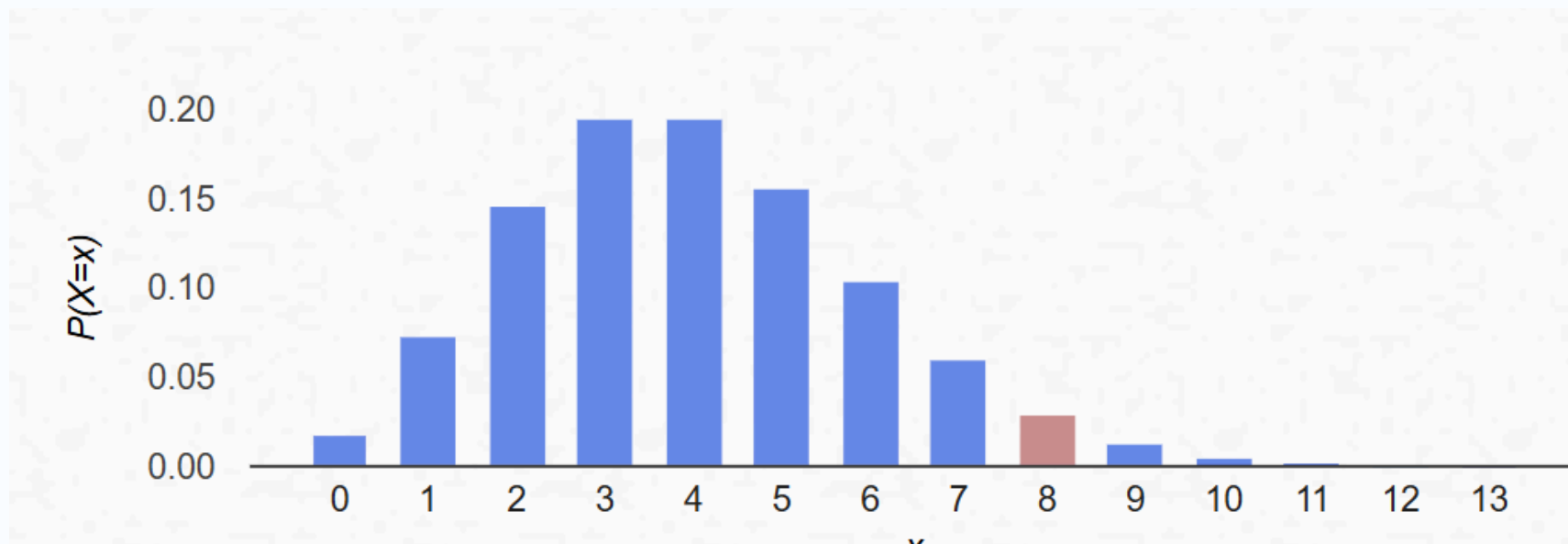
WHAT IS? WHAT IS THE CHARACTERISTICS? WHEN TO
USE? AND EXAMPLE TO USE IT

GLM POISSON

WHAT IS POISSON DISTRIBUTION?

1. “DISTRIBUSI UNTUK JUMLAH KEJADIAN DALAM SUATU INTERVAL (WAKTU/RUANG)
2. NOTASI SEDERHANA: λ = RATA-RATA KEJADIAN PER INTERVAL”
3. CIRI UTAMA: RATA-RATA $\approx$ VARIANS

$$\frac{\lambda^k e^{-\lambda}}{k!}$$



WHAT IS GLM POISSON

“REGRESI UNTUK DATA HITUNGAN (0,1,2,...) DENGAN LOG LINK.”

$$\log(\mathbb{E}[Y|X]) = \beta_0 + \beta_1 x_1 + \cdots + \beta_p x_p \Rightarrow \mathbb{E}[Y|X] = \exp(\beta_0 + \beta^\top X)$$

SEDERHANANYA MENJADI SEPERT

$$\hat{Y}_i = \hat{\lambda}_i = \exp\{\hat{\beta}_0 + \hat{\beta}_1 x_{i,1} + \cdots + \hat{\beta}_p x_{i,p}\}$$

- * i : indeks observasi (baris data), $i = 1, \dots, n$.
- * p : jumlah prediktor (fitur) dalam model.
- * $\hat{Y}_i$: prediksi ****rata-rata**** (expected count) untuk observasi ke- i .
- * $\hat{\lambda}_i$: taksiran parameter laju (mean Poisson) untuk observasi ke- i ; sama dengan $\hat{Y}_i$.
- * $\exp(\cdot)$: fungsi eksponensial; memastikan $\hat{\lambda}_i > 0$.
- * $\hat{\beta}_0$: ****intercept**** (baseline log-mean) saat semua prediktor bernilai 0.
- * $\hat{\beta}_j$: koefisien untuk prediktor ke- j ($j = 1, \dots, p$); $e^{\hat{\beta}_j}$ = faktor pengali pada mean untuk kenaikan 1 unit di $x_{i,j}$.
- * $x_{i,j}$: nilai prediktor ke- j pada observasi ke- i .

WHAT IS THE CHARACTERISTICS?

- 1.TARGET: BILANGAN BULAT ≥ 0 , SERING RIGHT-SKEWED (BANYAK NILAI KECIL, SEDIKIT NILAI BESAR)
- 2.ATURAN PRAKTIS POISSON: VARIANCE $\approx$ MEAN.
- 3.OFFSET (EXPOSURE) UNTUK BEDA DURASI/POPULASI:

$$\log(\mu) = \beta_0 + \beta^\top X + \log(\text{exposure}) \Rightarrow \text{model}$$

WHEN TO USE

PAKAI SAAT: TARGET = JUMLAH PER INTERVAL, PERLU INTERPRETASI PENGALI (IRR), ADA EXPOSURE BERBEDA.

EXAMPLE

$$\log(\mu) = \beta_0 + 0.35 \cdot \text{Promo} + 0.20 \cdot \text{Weekend}.$$

TARGET: JUMLAH PESANAN/HARI. FITUR: PROMO (0/1), WEEKEND (0/1).

Promo ON $\rightarrow \text{mean} \times e^{0.35} \approx 1.42$ (**+42%**).

Weekend $\rightarrow \text{mean} \times e^{0.20} \approx 1.22$ (**+22%**).

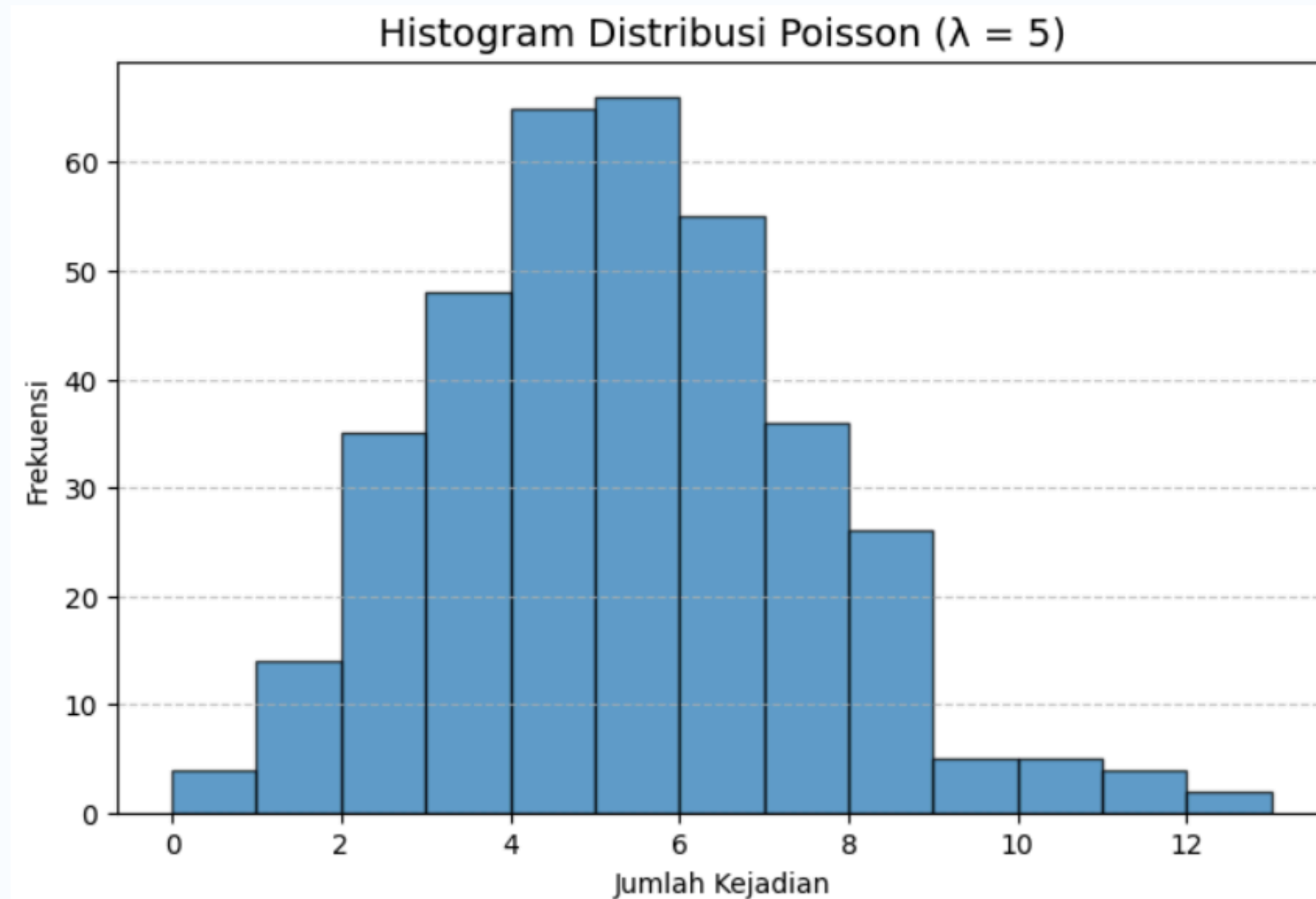
Predicting Daily Motorbike Credit Using Poisson GLM

365 DAYS OF DATA OBSERVATION GIVES A FULL-YEAR VIEW OF TRENDS AND VARIABILITY.

Factors affecting credit:

- *Weekend vs. weekday*
- *Promotional offers*

CHECK DISTRIBUTION: VARIANCE & MEAN



	Hari	Jumlah_Kredit
0	1	7
1	2	5
2	3	4
3	4	8
4	5	4

Statistik Distribusi Kredit Motor:
Rata-rata kredit motor (Mean): 4.94
Varians kredit motor: 5.07

POISSON ASSUMPTION: MEAN $\approx$ VARIANCE

ADD VARIABLE & SPLIT DATA

	Hari	Jumlah_Kredit	Weekend	Promo
0	1	7	0	0
1	2	5	0	1
2	3	4	0	1
3	4	8	0	0
4	5	4	0	0
5	6	8	1	0
6	7	7	0	0
7	8	5	0	1
8	9	9	0	0
9	10	4	0	1

```
Jumlah data total: 365
Jumlah data train: 276
Jumlah data test internal: 70
Jumlah data validasi: 19
```

- 95% UNTUK TRAIN+TEST INTERNAL (80% TRAIN, 20% TEST)
- 5% VALIDATION RESERVED FOR LATER EVALUATION

POISSON GLM MODEL

```
from sklearn.linear_model import PoissonRegressor

X_train = train_df[["Weekend", "Promo"]]
y_train = train_df["Credit_Count"]

model = PoissonRegressor(alpha=0, max_iter=300)
model.fit(X_train, y_train)
```

=== Koefisien Model ===

	Prediktor	Koefisien	Exp(Koef)
0	Intercept	1.569599	4.804721
1	Weekend	0.056314	1.057930
2	Promo	0.034526	1.035129

PREDICTION & EVALUATION

MSE Train: 5.183

MSE Test Internal: 4.705

Contoh 5 baris prediksi Train:

	Hari	Jumlah_Kredit	Prediksi	Prediksi_Rounded
116	213	1	4.770606	5
7	207	8	4.770606	5
153	257	5	4.926330	5
101	343	7	4.770606	5
307	277	6	4.926330	5

Contoh 5 baris prediksi Test Internal:

	Hari	Jumlah_Kredit	Prediksi	Prediksi_Rounded
108	170	8	4.770606	5
158	86	7	4.770606	5
278	132	3	4.894415	5
209	81	4	4.926330	5
320	1	7	4.770606	5

VALIDATION MODEL PREDICT

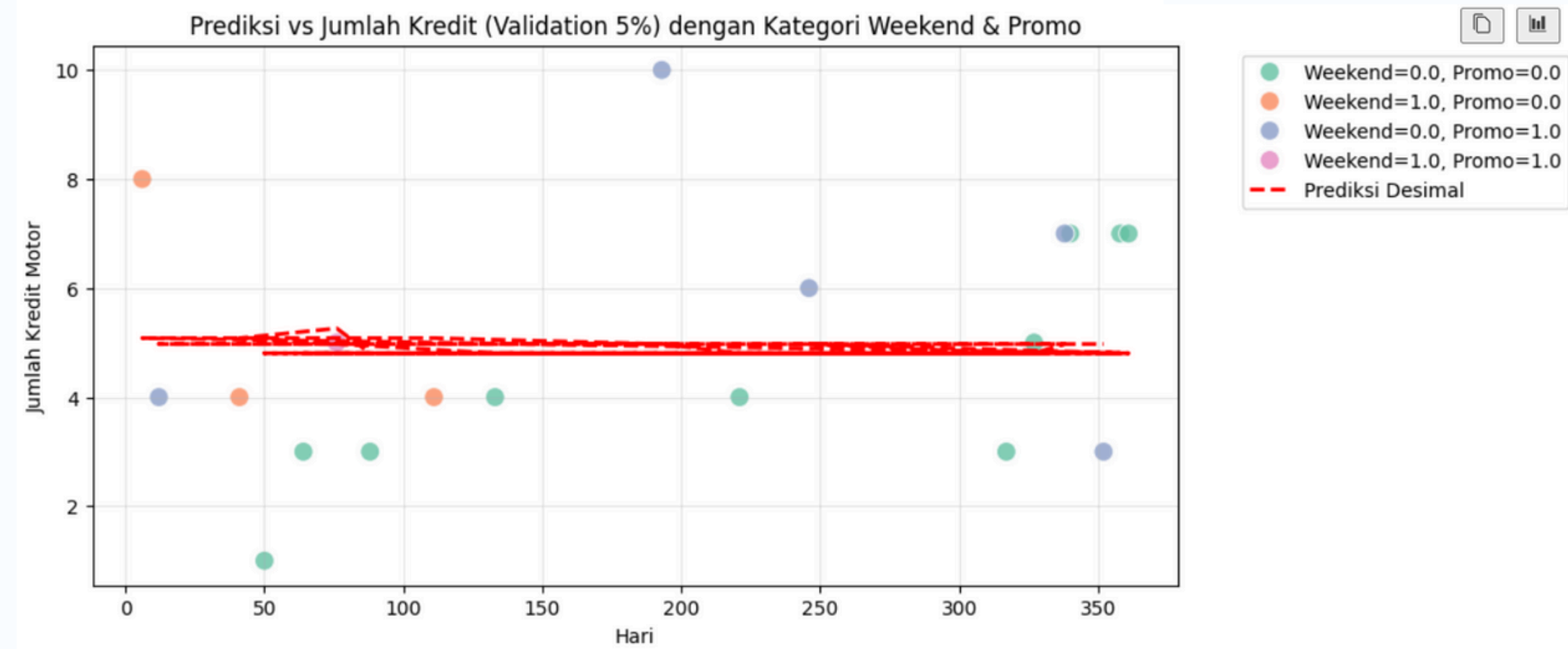
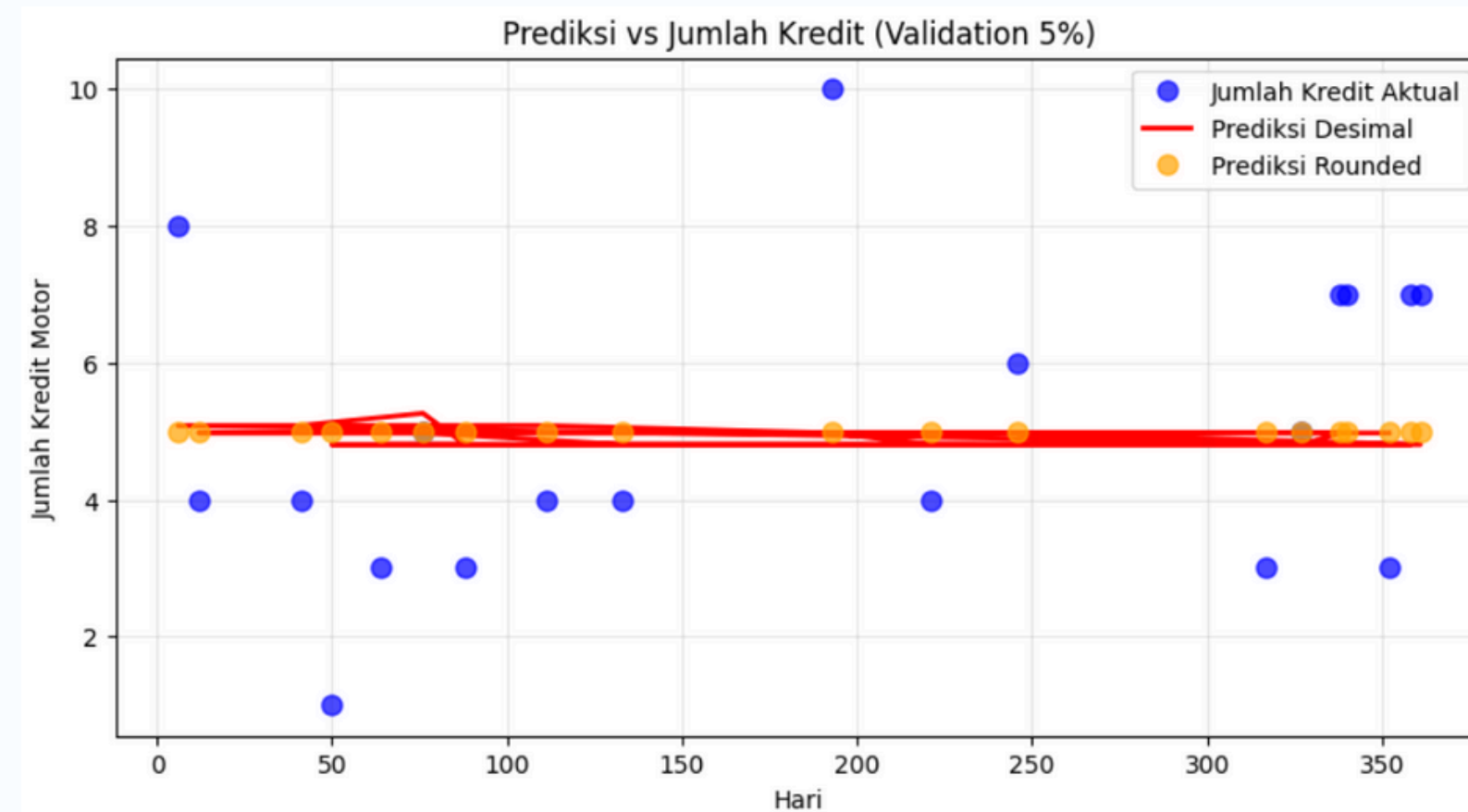
MSE Validation: 4.538

R² Validation: 0.020

Contoh 5 baris prediksi Validation:

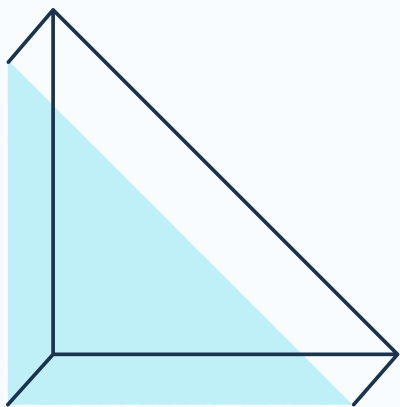
	Hari	Jumlah_Kredit	Prediksi	Prediksi_Rounded
346	340	7	4.804721	5
347	50	1	4.804721	5
348	358	7	4.804721	5
349	6	8	5.083057	5
350	111	4	5.083057	5
351	193	10	4.973505	5
352	221	4	4.804721	5
353	317	3	4.804721	5
354	246	6	4.973505	5
355	361	7	4.804721	5

VALIDATION MODEL PREDICT



Conclusion

- Poisson GLM successfully trained for predicting daily credit
 - Both Promo and Weekend positively contribute, but effects are very small
 - Predictions can guide inventory planning or promotional strategies



THANK YOU

"In a world of counts
and events, **Poisson**
shows us the
expected path."

Cetjipiti

